

THIRD SEMESTER, NOVEMBER 2024

Max. Marks: 30

Q. No.	Question	Max. Marks	CO(s)
1. (a)	Express the following sentence using quantifiers: “There is a natural number greater than 5.”	2	CO1
(b)	Let $A = \{a, b, c, d\}$, $B = \{c, d, e, f\}$. Find the set difference between $A - B$ and $B - A$, and determine the symmetric difference.	2	CO1
(c)	Explain the Universe of Discourse with an illustrative example.	2	CO1
(d)	Define a partially ordered set (poset) and explain the conditions that must be satisfied to be considered a poset.	2	CO2
(e)	Define the terms: Supremum and Infimum .	2	CO2
2. (a)	Demonstrate that R is a valid inference from the given set of premises: $P \rightarrow Q$, $Q \rightarrow R$ and P	5	CO1
(b)	Explain Reflexive closure , Symmetric closure , and Transitive closure with an example.	5	CO1
3. (a)	Simplify the following Boolean function using a 4-variable K-map: $F(A, B, C, D) = \Sigma(0, 1, 2, 5, 8, 9, 10, 14)$	5	CO2
(b)	Solve the recurrence relation $a_n = 7a_{n-1} - 10a_{n-2}$ with $a_0 = 2$ and $a_1 = 3$.	5	CO2
4. (a)	Draw Hasse Diagram of: $D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$ where $D_n = \{x : x \mid n, \text{ for all } x \in \mathbf{N}\}$. Check whether $(D_{30},)$ is a Lattice.	5	CO2
(b)	Consider set X as a set of natural numbers ($\mathbf{N}$), x, y are elements of X . Check whether the relation: $R = \{ (x, y) \in X \times X \mid x + y \text{ is even} \}$ is an Equivalence Relation. Justify for reflexivity, symmetry, and transitivity.	5	CO1